AgentCamp
Christchurch, New Zealand

Moving from “Block It” to Build It More Safely

Securing AI Agents in Microsoft Environments

Power BI and Fabric

SF
Steve Fox
Governance & Security
AnalyticsinAction.com

This space moves fast. Content reflects thinking as of March 2026.

Agenda

What we'll cover in this session

1. **The AI Landscape**
   Agentic tools operating in Power BI and Fabric
   Copilot, MCPs, coding agents, personal LLMs

2. **Key Risks**
   The most important threats to manage today
   Agency, misinformation, leakage, costs

3. **Guiding Principles**
   A practical governance framework for AI agents
   Educate, isolate, limit, recover

4. **What To Do Now**
   Actions you can take immediately
   Human, automated, and technical controls

The AI Landscape

AI-assisted development has reached a tipping point

1. **AI inside Fabric & Power BI**
   Copilot, Fabric Data Agents
   Native AI - governed by Microsoft

2. **MCPs - Connectors**
   Tabular Editor, Fabric Analytics, Power BI Modeling
   External tools connecting via MCP

3. **Agentic Coding Tools**
   Claude Code, Cursor, GitHub Copilot, Codex
   AI agents that write and execute code

4. **Personal LLMs**
   ChatGPT, Claude, Gemini
   General-purpose AI - hardest to track

If you're not seeing agents on your tenant, it's because tracking most AI usage is nearly impossible.

Risks

Can co-exist

OWASP LLM 2025

LLM Risks

1. **Prompt Injection**
   Malicious inputs that hijack AI behavior
2. **Naivety**
   Blindly trusting AI-generated outputs
3. **Man in the Middle**
   Intercepting AI communication channels
4. **Prompt Leakages**
   Exposing system instructions to users
5. **Info Stuffing**
   False information in information spare areas
6. **Supply Chain Hacks**
   Compromised models, plugins, or data

Today's Focus

Most relevant to Power BI & Fabric governance

7. **Excessive Agency**
   AI agents acting beyond intended scope, executing unauthorized actions without proper human oversight
8. **Sensitive Information Access**
   AI models exposing confidential data, PII, or proprietary information through responses or retrieval
9. **Hallucinations**
   AI generating plausible but incorrect information, leading to flawed decisions based on fabricated data
10. **Unbound Consumption**
    Unconstrained resource usage by AI agents causing excessive compute costs or capacity exhaustion

Arguably these 4 risks are the most relevant to **Power BI** and **Fabric Security** governance today.

Guiding Principles

Educate, Isolate, Limit Privileges, Ensure Recoverability

EDUCATE

- Risk continuum (Vendors Agents vs OpenClaw)
- Encourage best practices
- Visibility via analyst groups
- Regular training to keep pace
- Mitigate risks (CI/CD etc.)

ISOLATION

- Keep agents away from Prod & sensitive data
- Separate environments (Dev, Test, Prod)
- Capacity isolation (F SKU)
- Dev Sandboxes: Containers, VDE
- Data masking

PRIVILEGE

- User with tenant admin = Agent with tenant admin
- Agent-specific identities (linked to users)
- Assign minimal reasonable permissions

RECOVERY

- If an agent breaks something, can you recover?
- CI/CD pipelines with automated tests
- Source control everything (Git / SharePoint)

Four pillars to secure AI agents in your **Fabric** environment

Agent Security Roadmap

Seven steps to secure your agentic environment

The gate is open – it will be easier to act before adoption accelerates

Resources & Deeper Dives

Scan to access all references

What You'll Find

1. Slides (PDF download)
2. Referenced articles
3. Reach me: linkedin.com/in/stevefoxnz/

Thank You!

Questions?

Scan for links & resources